

Standard operating procedure for the cells

Three parametric cells, and every parameter of each one: what it is, what moves when you move it, and where its default came from.

That last column is the reason this document exists. A PDK looks authoritative whether or not its numbers are: a default of 5 um reads the same whether it was measured on silicon, read off a drawing, or picked because it seemed reasonable. Here they are labelled.

measured	measured on this process, on these chips
off the chip	read off the fabricated test chip's GDS
ASSUMED	nobody has measured this - treat any result that leans on it as provisional
your choice	a design decision; nothing about the process fixes it

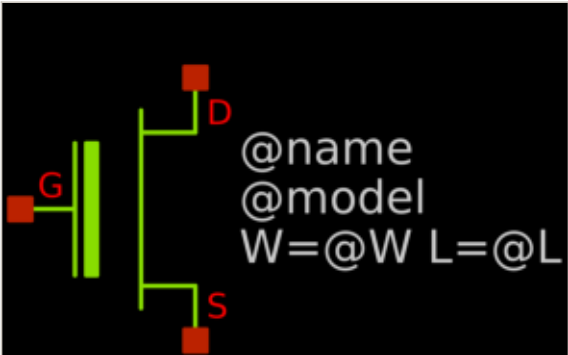
Placing a cell

In KLayout: Instance (I), library igzo_mmm_lab_pr, one of tft_igzo, cap_mim, ind_igzo. The parameters below appear in the instance dialog; the read-only ones update as you type.
In xschem: the symbols are on the library path once use-pdk has run.
Both are described cell by cell in the pages that follow.

tft_igzo - the transistor

Three terminals, D G S, in that order. There is no bulk: a TFT sits on glass, and the wrapper subcircuit keeps the intrinsic device's bulk tied to its own source.

The defaults reproduce TFT-10L100W of the test chip shape for shape - so an unmodified tft_igzo is a device that has actually been fabricated and measured, not a guess at one.



Left: the drawn device with its probe pads. Right: the symbol.

tft_igzo - the transistor

Parameter	Default	What it does	Source
w_gate	100 um	channel width per finger	your choice
l_gate	10 um	drawn channel length = electrode gap	your choice
nf	1	fingers; they share electrodes	your choice
gate_ov	5 um	gate overlap per side - the dominant parasitic	off the chip
sd_len	97.5 um	S/D electrode length	off the chip
igzo_ext_x	10 um	island over each electrode	off the chip
igzo_ext_y	5 um	island past the electrode end	off the chip
gate_ext_y	10 um	gate past the island	off the chip
pad	off	400 x 400 um probe pads + dielectric windows	your choice
pad_size	400 um	pad side	off the chip
pad_gap	325 um	clearance between S/D pads and gate pad	off the chip
etch_inset	20 um	window inset inside the pad	off the chip
l_bias	3 um	drawn-to-printed bias, reported not applied	ASSUMED
l_printed	read-only	l_gate + l_bias	-
w_total	read-only	w_gate x nf	-

Two things worth knowing before you size one

FINGERS DO NOTHING ELECTRICALLY. nf splits the width and folds the layout; gm and Cgs split with it, so their ratio - and fT - do not move. Measured, then confirmed in simulation for nf = 1..20 at constant total width. Use fingers for aspect ratio and gate resistance, not for speed.

L IS THE DRAWN LENGTH. The process prints it about 3 um longer. The cell reports l_printed and leaves the geometry alone, because that +3 um is one measurement of one cell: applying it silently would assert more than anyone knows. If you want the printed device, pass the printed L.

The model side of the same device

In a netlist the device is a subcircuit call with its own parameters, which are not quite the PCell's:

Netlist parameter	Meaning
W, L	drawn width and length, in metres
ov	gate overlap per side; sets the overlap capacitance, $C_{ox} \cdot ov \cdot W$
nf	carried for LVS only - it changes nothing electrically
B	magnetic field in tesla, default 0 - see below
b_scale	dimensionless, default 1: how much larger than classical
corner	chosen by the .lib line, not by the instance

Three terminals, and no wrong way round

d, g, s - there is no bulk pin, and the intrinsic device's substrate node reaches nothing: every junction on every model card is inert. So the device is symmetric, and swapping d and s changes no number anywhere. Draw the stage the way it reads best.

B carries classical magnetoresistance, $\mu / (1 + (\mu \cdot b_scale \cdot B)^2)$, with μ from the corner's own K_p / C_{ox} . At our mobility that is $2e-7$ per tesla - present, honest, and unmeasurable. docs/bibliography/ says why, and what to set b_scale to when a real coefficient has been measured.

cap_mim - the capacitor

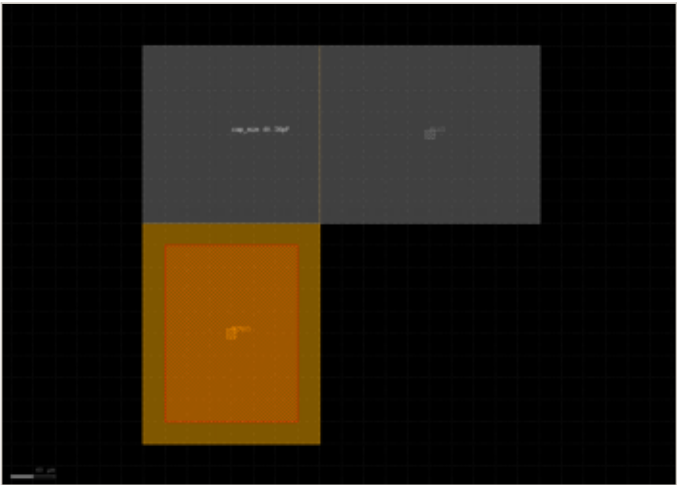
Set the plate, or the area, or the capacitance - whichever the problem hands you. It is one formula solved in three directions:

$C = C_{ox} * W * L$, $C_{ox} = 1.564 \text{ fF/um}^2$ (measured)

The two you did not set are read back, and the cell prints the result on itself on layer 63/0, which is annotation and never fabricated.

mode	you set	you get
by dimensions	w = l = 400 um	area 160000 um^2, C = 250.2 pF
by area	area = 160000 um^2	w = l = 400 um, C = 250.2 pF
by capacitance	c = 40362 fF	w = l = 160.64 um, area 25805 um^2

Parameter	Default	What it does	Source
mode	by dimensions	which of W/L, area or C you set	your choice
w, l	400 um	plate, used in 'by dimensions'	your choice
area_target	160000 um^2	used in 'by area' - square plate	your choice
c_target	250240 fF	used in 'by capacitance' - square plate	your choice
ext_sd	200 um	S/D plate past the crossing, terminal side	your choice
ext_gate	200 um	gate plate past the crossing, terminal side	your choice
ext_sd_far	0	S/D plate past the crossing, opposite side	your choice
ext_gate_far	0	gate plate past the crossing, opposite side	your choice
pad	off	probe pad on the end of each plate	your choice
pad_size	400 um	pad side	off the chip
etch_inset	20 um	window inset in the lower plate's tail	off the chip
show_value	on	print the capacitance on layer 63/0	your choice
area, c	read-only	always $C_{ox} * W * L$, $C_{ox} = 1.564 \text{ fF/um}^2$	measured

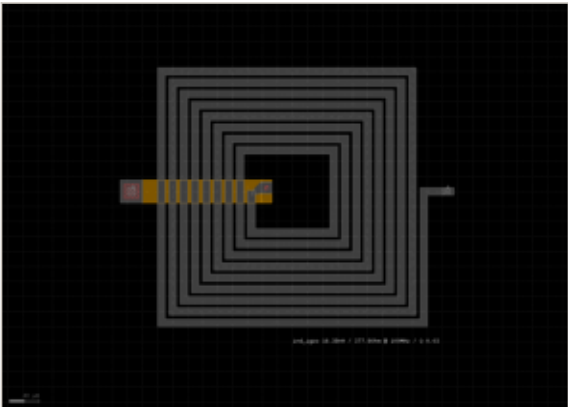


cap_mim sized by capacitance: 40.36 pF, and it says so on itself.

ind_igzo - the inductor

Square or circular, and either one spiral in series or n concentric rings in parallel. The estimates come from tools/coil_core.py - the same library as the coil calculator, so the two always agree - and the resistance is computed from the length of track actually drawn, leads included, rather than from a closed form for it.

Parameter	Default	What it does	Source
shape	square	square or circular	your choice
topology	series	one spiral, or n rings in parallel	your choice
n	8	turns (series) or rings (parallel)	your choice
d_in	100 um	inner diameter	your choice
w	10 um	track width - sets R, barely touches L	your choice
gap	5 um	spacing between turns	your choice
metal	gate (2/0)	which gold to draw it on	your choice
t_metal	0.05 um	metal thickness - the lever on R	ASSUMED
f_eval	100 MHz	frequency the R_ac and Q read-outs use	your choice
lead	40 um	terminal lead length	your choice
pad	off	probe pad on each terminal	your choice
pad_size	200 um	pad side	your choice
lbl	on	pins and labels A and B - needed for extraction	your choice
l_series	read-only	Mohan-Wheeler (series) or coupled rings	-
r_dc, r_ac	read-only	from the drawn track length and rho_Au	-
q, d_out, wire_len	read-only	Q at f_eval, outer size, track length	-

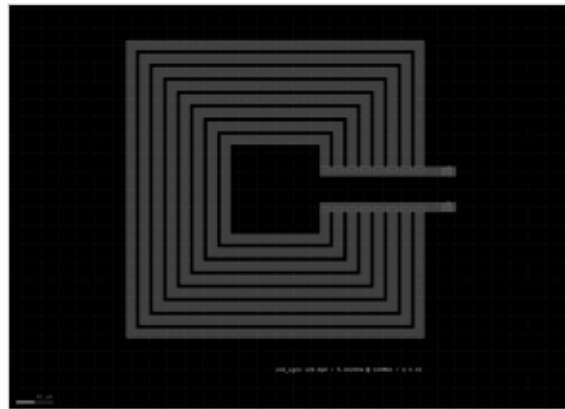


Series spiral: the turns on gate metal, the inner end taken out underneath on S/D metal through two openings in the dielectric.

ind_igzo - the inductor

shape / topology	L	R at 100 MHz	Q
square, series	16.38 nH	377.7 Ohm	0.03
circular, series	20.59 nH	312.6 Ohm	0.04
square, parallel	129.6 pH	5.00 Ohm	0.02
circular, parallel	129.5 pH	3.87 Ohm	0.02

All four at the default size, $n = 8$, 50 nm of gold. Parallel rings buy a low resistance and almost no inductance - they are a current path, not a tank element. A circular spiral gives about 25 % more L than a square one of the same inner diameter, and slightly less resistance.



Parallel rings: cut at the same angle, both buses running out through the slot. No via needed - nothing crosses.

The number that decides whether any of this is usable

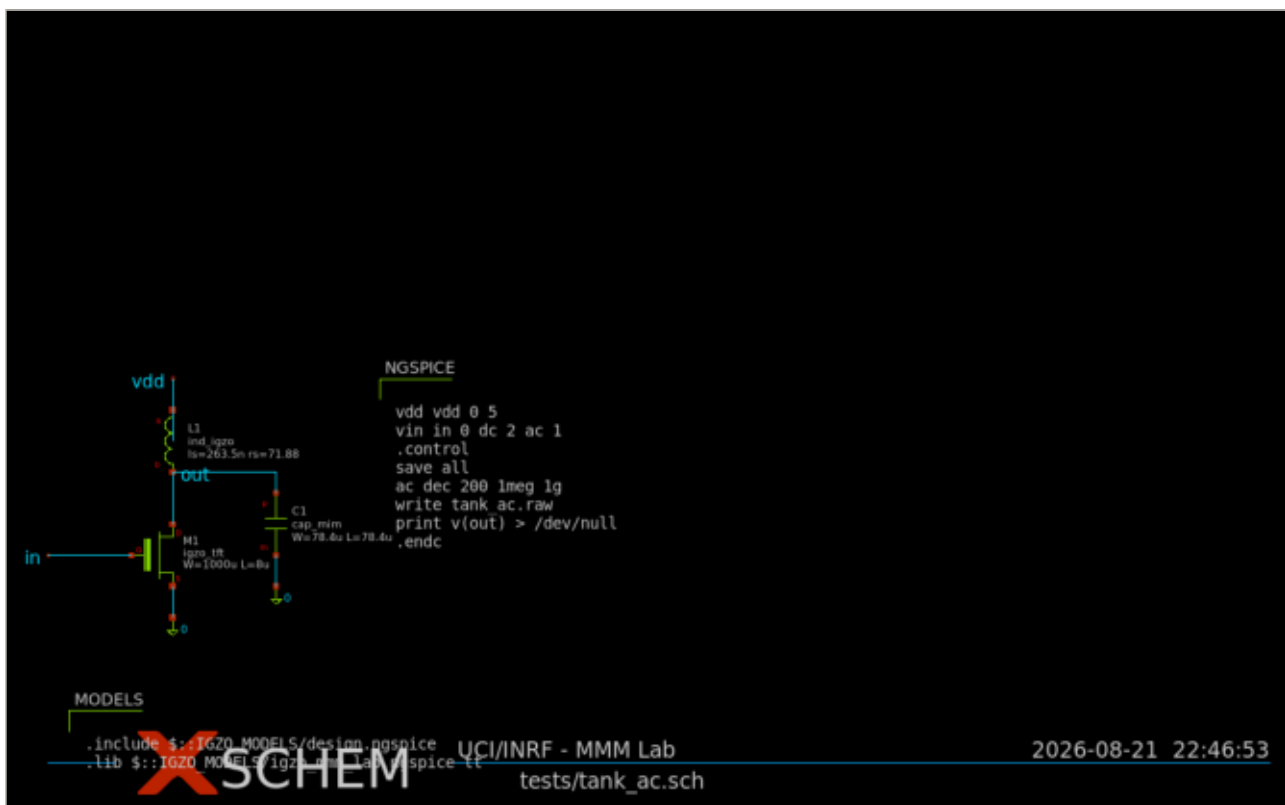
gold thickness	R at 100 MHz	Q
50 nm (assumed today)	377.7 Ohm	0.03
100 nm	188.9 Ohm	0.05
250 nm	75.5 Ohm	0.14
500 nm	37.8 Ohm	0.27
1 um	18.9 Ohm	0.54

The number that decides whether any of this is usable

Same coil, same inductance, thickness swept. At the 50 nm the process is assumed to have, a planar inductor here is not a tank element - it is a resistor with some inductance. Q scales with thickness and with nothing else you can draw.

The 100 MHz tank in tests/tank_ac.sch shows the same thing end to end: 263.5 nH with 9.61 pF resonates at 100.0 MHz in 1 μm of gold, and in 50 nm the peak does not exist at all.

And the thickness itself is assumed. Until somebody measures the gold, every resistance and every Q in this PDK is provisional - which is why t_{metal} is a parameter on the cell and not a constant inside it.



tests/tank_ac.sch - the three cells together.

Extraction, parasitics and LVS

Can a netlist be extracted from layout, and LVS'd with netgen?

Yes, and both are checked on every run of scripts/run_checks.sh.

```
magic -rcfile $PDKPATH/libs.tech/magic/igzo_mmm_lab.magicrc
load <cell> ; extract all ; ext2spice lvs ; ext2spice
netgen -batch lvs "layout.spice cell" "schematic.spice cell" \
    $PDKPATH/libs.tech/netgen/setup.tcl
```

The transistor extracts as an igzo_tft subcircuit with the L you drew and pins in D G S order, and LVS against the xschem netlist matches uniquely. The capacitor extracts as a capacitance between its two ports. The inductor does not extract as a device at all - see below.

Can parasitic capacitance be extracted?

Yes, and it is exact for this process because there is only one dielectric to know about:

```
ext2spice cthresh 0      (0 = keep every capacitance)
```

Drawn 40.36 pF, extracted 40.3593 pF; drawn 250.24 pF, extracted 0.25024 nF. The same number appears wherever gate metal happens to run over S/D metal in a design, which is the parasitic worth having: at 1.564 fF/um² a crossing is not free.

There is no substrate capacitance in this technology and the PDK does not invent one. A TFT sits on glass; there is nothing underneath to couple to.

Extraction, parasitics and LVS

Can parasitic resistance be extracted?

Partly, and this is where the honest answer is longer than yes.

The techfile carries the sheet resistance - 488 mOhm/square, which is gold at the assumed 50 nm - so anything that uses it gets the right number, and run_checks.sh asserts that it agrees with what the coil cell reports: $774 \text{ squares} \times 0.488 = 377.7 \text{ Ohm}$, which is what ind_igzo prints on itself. Change the thickness and both move together.

What does not work is magic's extresist pipeline. On a coil it warns that the two ports are electrically shorted - which is true, a coil is a short at DC - and then crashes; on simpler shapes it produces no output. So interconnect resistance in this PDK comes from the cell estimate and from the sheet resistance, not from extresist. If you need it per-net on a real block, the route is $\text{squares} \times 0.488 \text{ Ohm}$ on the geometry, not magic.

What this means for an inductor in LVS

A coil extracts as one net: its two terminals are the same piece of metal. Nothing in the layout says 'inductor', because nothing in the layout is one - it is a wire whose shape happens to store energy.

So an ind_igzo instance in a schematic will not match an extracted coil pin for pin. Treat the coil as a black box for LVS: keep it as a fixed cell, compare the block around it, and check the coil itself by geometry - which is what the connectivity check in run_checks.sh does, after an inner ring was once left floating by a bus that started in the wrong place and nothing noticed.

The layers, and the one rule that changed

Four masks, in process order: igzo (4/0), sd (6/0), oxetch (5/0), gate (2/0), plus align (8/0) and text (63/0), which is annotation.

oxetch does two jobs. Over a pad with nothing above it, it is a probe window. Under gate metal it is a VIA: the gate is deposited after the etch, so the two metals touch. That is what lets a spiral inductor bring its inner end out, and it is why the DRC deck has via rules where it used to have a rule forbidding gate over an opening.

Rule	Value	What it protects
IGZO.1 / SD.1 / GATE.1	5 um	minimum drawn width of anything
SD.2	5 um	S/D spacing - this is the drawn channel length
GATE.2 / IGZO.3	5 um	spacing on gate and on the island
GATE.3	5 um	gate overlap of the electrode, where there is a channel
IGZO.2	5 um	S/D contact overlap on the IGZO
OX.1 / OX.2	- / 20 um	openings land on metal; probe window inset
VIA.1 / VIA.2 / VIA.3	5 um	via size, and both metals' enclosure of it

Before you release anything

1. DRC: `python3 libs.tech/klayout/tech/drc/run_drc.py --path x.gds`
2. Extract in magic and read the netlist: the device should come out as `igzo_tft` with the L you drew. W will read about 13 % high, because the extractor measures the gated island and the island is drawn past the electrodes for alignment margin.
3. LVS through `libs.tech/netgen/setup.tcl`, which allows for exactly that difference and holds L to 1 %.
4. `./scripts/run_checks.sh` - the whole PDK, twelve checks, end to end.

One known residual: the circular series inductor leaves seven violations where its inner terminal meets the first turn tangentially. Every other cell in the library is clean.